

# Experiment 21: Towers of Hanoi Using Prolog

## Aim

To implement the **Towers of Hanoi problem using Prolog** and demonstrate the use of recursion to move disks from one peg to another.

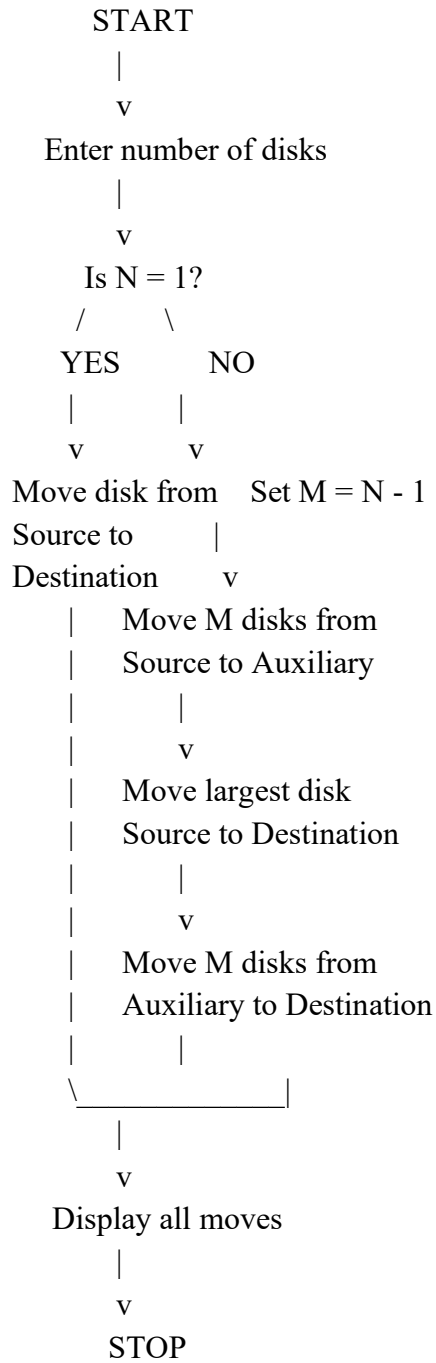
## Objective

1. To understand recursion in Prolog.
2. To implement the Towers of Hanoi problem using Prolog rules.
3. To generate the sequence of moves required to transfer disks.
4. To understand how Prolog performs logical inference.

## Algorithm

1. Start the program.
2. Define the base case for **one disk**.
3. For  $N > 1$ , calculate  $N-1$ .
4. Move  $N-1$  disks from the **source peg to the auxiliary peg**.
5. Move the remaining largest disk from the **source peg to the destination peg**.
6. Move  $N-1$  disks from the **auxiliary peg to the destination peg**.
7. Repeat the process recursively until all disks are moved.
8. Display each move.
9. Stop.

# Flowchart



## Prolog Code

% Towers of Hanoi

```
hanoi(1, Source, Destination, _) :-  
    write('Move disk from '),  
    write(Source),  
    write(' to '),  
    write(Destination),  
    nl.
```

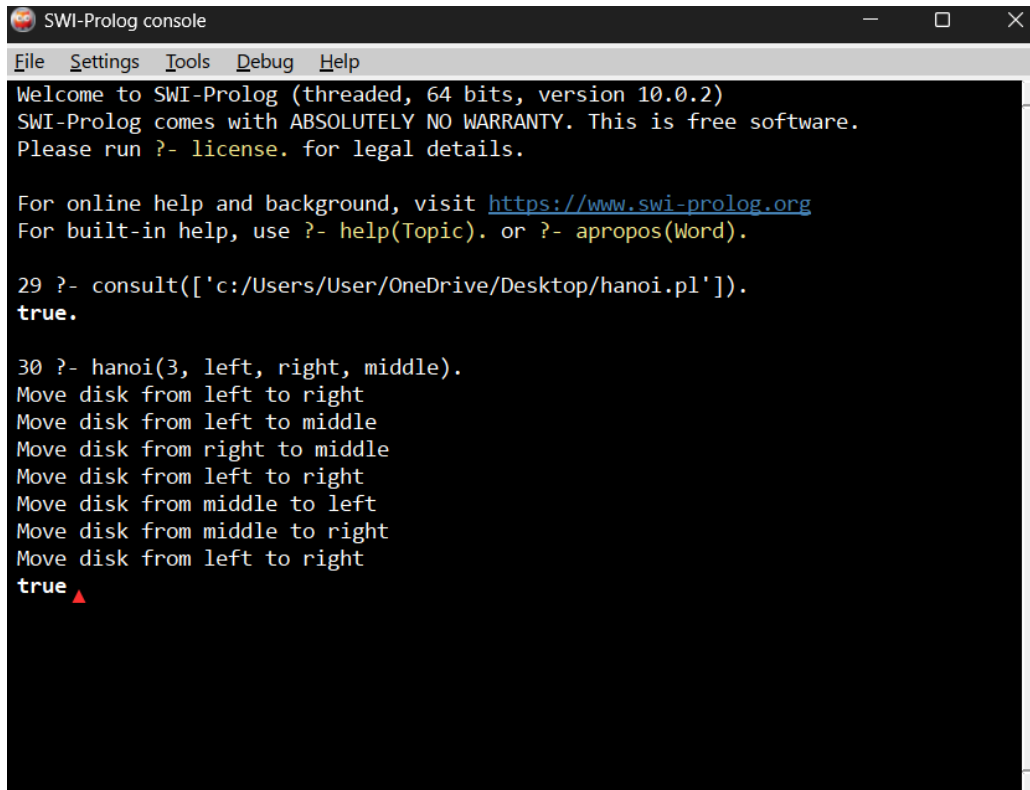
```
hanoi(N, Source, Destination, Auxiliary) :-  
    N > 1,  
    M is N - 1,  
    hanoi(M, Source, Auxiliary, Destination),  
    hanoi(1, Source, Destination, Auxiliary),  
    hanoi(M, Auxiliary, Destination, Source).
```

## Sample Query

For **3 disks**, execute:

```
?- hanoi(3, left, right, middle).
```

## Sample Output



```
SWI-Prolog console
File Settings Tools Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

29 ?- consult(['c:/Users/User/OneDrive/Desktop/hanoi.pl']).
true.

30 ?- hanoi(3, left, right, middle).
Move disk from left to right
Move disk from left to middle
Move disk from right to middle
Move disk from left to right
Move disk from middle to left
Move disk from middle to right
Move disk from left to right
true ▲
```

## Result

The **Towers of Hanoi problem** was **successfully implemented using Prolog**. The program correctly generated the required sequence of moves for transferring the disks from the source peg to the destination peg.

## Conclusion

Thus, the experiment demonstrates the use of **recursion in Prolog** to solve the Towers of Hanoi problem. The program efficiently generates the minimum sequence of moves required to transfer all disks while following the rules of the Towers of Hanoi.